

1) a) $(4195.25)_{10}$

$$\begin{array}{r}
 2 \overline{) 4195} \\
 \underline{2097} - 1 \\
 2 \overline{) 1048} - 1 \\
 \underline{524} - 0 \\
 2 \overline{) 262} - 0 \\
 \underline{131} - 0 \\
 2 \overline{) 65} - 1 \\
 \underline{32} - 1 \\
 2 \overline{) 16} - 0 \\
 \underline{8} - 0 \\
 2 \overline{) 4} - 0 \\
 \underline{2} - 0 \\
 2 \overline{) 1} - 0 \\
 \underline{0} - 1
 \end{array}$$

$$= (1000001100011)_2$$

b) $(2356.54)_{10}$

$$\begin{array}{r}
 2 \overline{) 2356} \\
 \underline{1178} - 0 \\
 2 \overline{) 589} - 0 \\
 \underline{294} - 1 \\
 2 \overline{) 147} - 0 \\
 \underline{73} - 1 \\
 2 \overline{) 36} - 1 \\
 \underline{18} - 0 \\
 2 \overline{) 9} - 0 \\
 \underline{4} - 1 \\
 2 \overline{) 2} - 0 \\
 \underline{1} - 0 \\
 \underline{0} - 1
 \end{array}$$

$$= (100100110100)_2$$

$$\begin{aligned}
 0.25 \times 2 &= 0.5 & 0 \\
 0.5 \times 2 &= 1.0 & 1
 \end{aligned}$$

$$(.25)_{10} = (.01)_2$$

$$\therefore (4195.25)_{10} = (1000001100011.01)_2$$

$$0.54 \times 2 = 1.08 \quad 1$$

$$1.08 \times 2 = 2.16 \quad 0$$

$$0.16 \times 2 = 0.32 \quad 0$$

$$0.32 \times 2 = 0.64 \quad 0$$

$$(.54)_{10} = (.1000)_2$$

$$\therefore (2356.54)_{10} = (100100110100.1000 \dots)_2$$

$$2) a) (A9)_{19} = (?)_7$$

$$= (A9)_{11} = A \times 11^1 + 9 \times 11^0 = 119$$

$$\begin{array}{r} 7 \overline{) 119} \\ 7 \overline{) 11} - 0 \\ 7 \overline{) 2} - 3 \\ 0 - 2 \end{array}$$

$$(119)_{10} = (230)_7$$

$$b) (0011)_{BCD} = (3)_{10} = (\text{3})_5$$

$$\begin{array}{r} 5 \overline{) 3} \\ 0 - 3 \end{array}$$

$$c) (1036)_{10} = (?)_{e-3}$$

$$= \begin{array}{cccc} 0001 & 0000 & 0011 & 0110 \\ +3 & +3 & +3 & +3 \end{array}$$

$$= (0100 \ 0011 \ 0110 \ 1001)_{e-3}$$

$$d) \underline{0111} \underline{1001} \underline{0011} \underline{1010} \underline{1000} \underline{1010} \underline{1100}$$

$$= (393 \cdot A8AC)_{16}$$

$$3) \quad (A3)_{16} \quad (47)_{16}$$

$$\begin{array}{r} A3 \\ + 47 \\ \hline EA \end{array}$$

$$\begin{array}{r} 9A3^{+16} \\ - 47 \\ \hline 5C \end{array}$$

$$\begin{array}{r} 2^4 1 A3 \\ \times 47 \\ \hline 1475 \\ 28C \times \\ \hline 2D35 \end{array}$$

$$(A3)_{16} = A \times 16^1 + 3 \times 16^0 = (163)_{10}$$

$$(47)_{16} = 4 \times 16^1 + 7 \times 16^0 = (71)_{10}$$

$$163 + 71 = (234)_{10}$$

$$163 - 71 = (92)_{10}$$

$$163 \times 71 = (11573)_{10}$$

$$(EA)_{16} = E \times 16^1 + A \times 16^0 = (234)_{10}$$

$$(5C)_{16} = 5 \times 16^1 + C \times 16^0 = (92)_{10}$$

$$(2D35)_{16} = 2 \times 16^3 + D \times 16^2 + 3 \times 16^1 + 5 \times 16^0 = (11573)_{10}$$

$$4) \quad 91 - 499 \\ = 91 + (-499)$$

$$\begin{array}{r} 2 \overline{) 91} \\ 2 \overline{) 45 - 1} \\ 2 \overline{) 22 - 1} \\ 2 \overline{) 11 - 0} \\ 2 \overline{) 5 - 1} \\ 2 \overline{) 2 - 1} \\ 2 \overline{) 1 - 0} \\ 6 - 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 499} \\ 2 \overline{) 249 - 1} \\ 2 \overline{) 124 - 1} \\ 2 \overline{) 62 - 0} \\ 2 \overline{) 31 - 0} \\ 2 \overline{) 15 - 1} \\ 2 \overline{) 7 - 1} \\ 2 \overline{) 3 - 1} \\ 2 \overline{) 1 - 1} \\ 0 - 1 \end{array}$$

$$499 = 111110011$$

$$+ 499 = 0111110011 \\ \text{in 10 bits}$$

$$- 499 = 1000001100 + 1$$

$$(1000001101)_{25}$$

$$(91)_{10} = 1011011$$

$$+ 91 = (0001011011)_{25}$$

in 10-bits

$$\begin{array}{r} 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ 0001011011 \\ + 1000001101 \\ \hline 1001101000 \end{array}$$

No overflow, since we are adding two different signed numbers.

$$5) a) (26)_8 \times (47)_8$$

$$\begin{array}{r} 5^1 3^2 2^6 \\ \times 4^7 \\ \hline 232 \\ + 130 \times \\ \hline (1532)_8 \end{array}$$

$$\begin{array}{r} 5 \\ 8 \sqrt{42} \\ -40 \\ \hline 2 \end{array}$$

$$\begin{array}{r} 2 \\ 8 \sqrt{19} \\ -16 \\ \hline 3 \end{array}$$

$$\begin{array}{r} 3 \\ 8 \sqrt{24} \\ -24 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 1 \\ 8 \sqrt{11} \\ -8 \\ \hline 3 \end{array}$$

$$b) \text{ Grant} = (4F0)_{16}$$

$$\text{temperature sensors} = (1532)_8$$

$$\begin{array}{l} \text{cost} \\ = (001101011010)_2 \\ = (35A)_{16} \end{array}$$

$$\begin{array}{l} \text{soil moisture} \\ \text{cost} \\ = (0010110010)_2 \\ = (172)_{16} \end{array}$$

$$\begin{array}{r} \text{Remaining money} = 4F0^{E+16} \\ 35A \\ -172 \\ \hline (024)_{16} \end{array}$$

$$c) \begin{array}{r} 3 \\ C \sqrt{24} \\ -24 \\ \hline 0 \end{array}$$

$$C \times 0 = 0$$

$$C \times 1 = C$$

$$C \times 2 = 18$$

$$C \times 3 = 24$$

$$\begin{array}{r} 1 \\ 16 \sqrt{24} \\ -16 \\ \hline 8 \end{array}$$

$$\begin{array}{r} 2 \\ 16 \sqrt{36} \\ -32 \\ \hline 4 \end{array}$$

$$\therefore \text{Bonus each engineer received} = (3)_{16}$$